

● **Group E**

Residential Energy consumption modelling



Introduction

Our task was to model how domestic energy usage has changed over time in the UK and forecast it for the next 5 years.

To achieve this we studied and analysed the last 35 years of government and public data on energy consumption and household income to create an accurate multiple regression model.

This is important because over the last 5 years energy prices have fluctuated wildly due to geopolitical uncertainty and conflict. These factors heavily affect the price of gas and electricity causing stress on the UK's energy infrastructure contributing to a cost of living crisis.

Therefore, this forecast is important because It will help shape the UK's energy policy allowing it to de-risk and ease the transition to renewable and sustainable sources.



Project Overview

Project Goals:

- Create a multiple regression model
- Create an accurate 5 year forecast
- Ensure our model is robust and has minimal multicollinearity
- Provide accurate policy and economic recommendations based on multiple different scenarios

Project Scope:

- The model is only domestic data so Industrial data is not included
- We have taken a top down approach to modelling data so seasonal fluctuations have not been analysed.
- We have only used data from reputable sources mainly ONS and Statista



Conceptual overview

Why we have chose the UK as our sustainable country:

- By living and studying in the UK, we are stakeholders in the UK energy system, allowing us to provide our own conceptual knowledge.
- The UK has a high availability of reliable data, from government sources (including ONS), and from Statista which is easily accessible to the public.
- The energy price shocks of 2021-2023 have provided shock within the UK energy market and provide reason for policy change to help the welfare off the consumer.
- The UK government has stated a policy of which they have set targets for net zero in 2050, which enables us to build a report informing policy makers on progress and necessary changes to be made in order to achieve this goal.



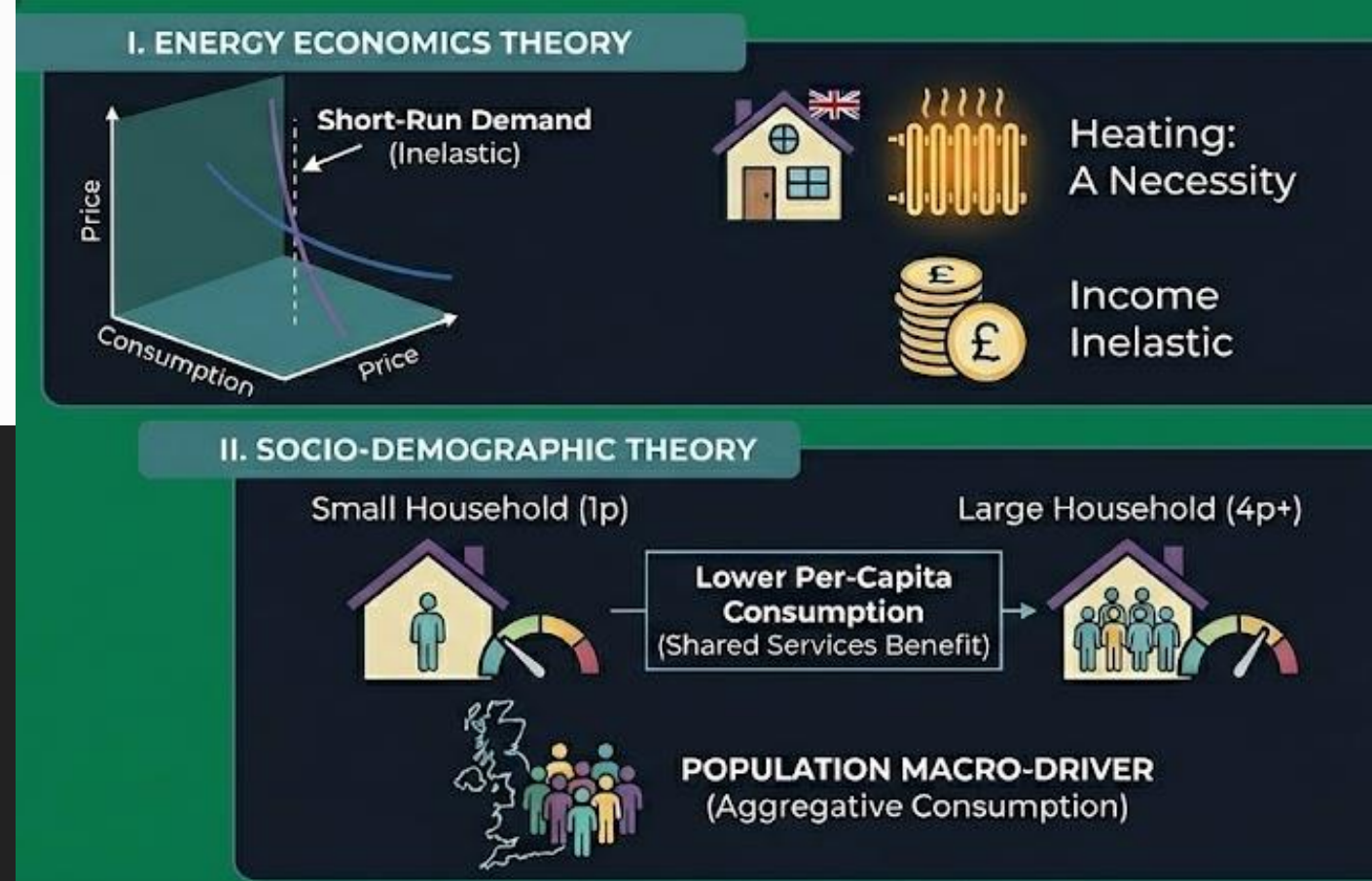
Conceptual overview continued...

Theoretical foundations:

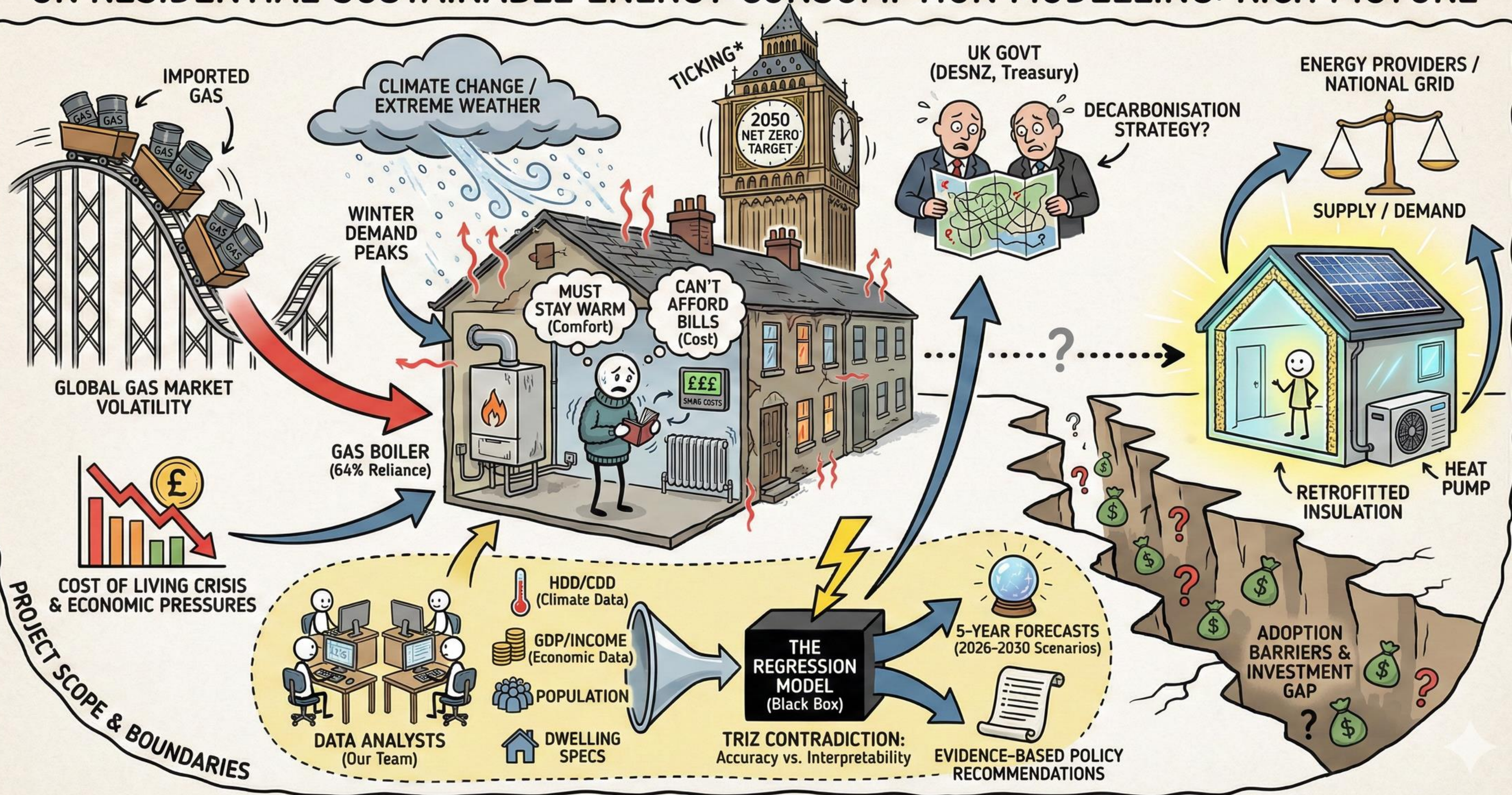
- Energy economics theory: In the short-run, domestic energy is typically price inelastic, and only becomes more elastic in the long-run. In the context of income-elasticity, energy tends to be income inelastic due to the majority of usage being used for heating which is a necessity for homes in the UK.
- Socio-demographic theory: Larger household size can result in a lower consumption per capita due to shared energy services. Overall UK population can drive overall aggregative consumption pointing it as a core macro-driver.

Conceptual foundation for multiple regression modelling:

- Isolates the marginal effect of each factor while controlling it for other inputs.
- Allows for model comparison and allows for testing effects of price changes.
- Supports ability for forecasting which allows to produce relevant recommendations for policymakers.



UK RESIDENTIAL SUSTAINABLE ENERGY CONSUMPTION MODELLING: RICH PICTURE



SSM: Root Definition + CATWOE

"A system owned by the UK Government, operated by data analysts, to transform inaccurate estimates into data-driven energy forecasts for UK households and policymakers. This assumes that proactive planning is essential for a sustainable future, while navigating the constraints of Net Zero targets and extreme price volatility."



Analytical Design

- Government Data and Statista - kept it from as few different sources as possible
- All data goes back to 1990
- Why two different Models?
- Why variables? Initial list provided, Other top down studies, own conceptual knowledge/common sense

Regression Model Process

1

Correlation

2

Initial Model

3

Refinement

4

Assumptions
What will happen if we change
one variable (electricity price)



Regression Model Output

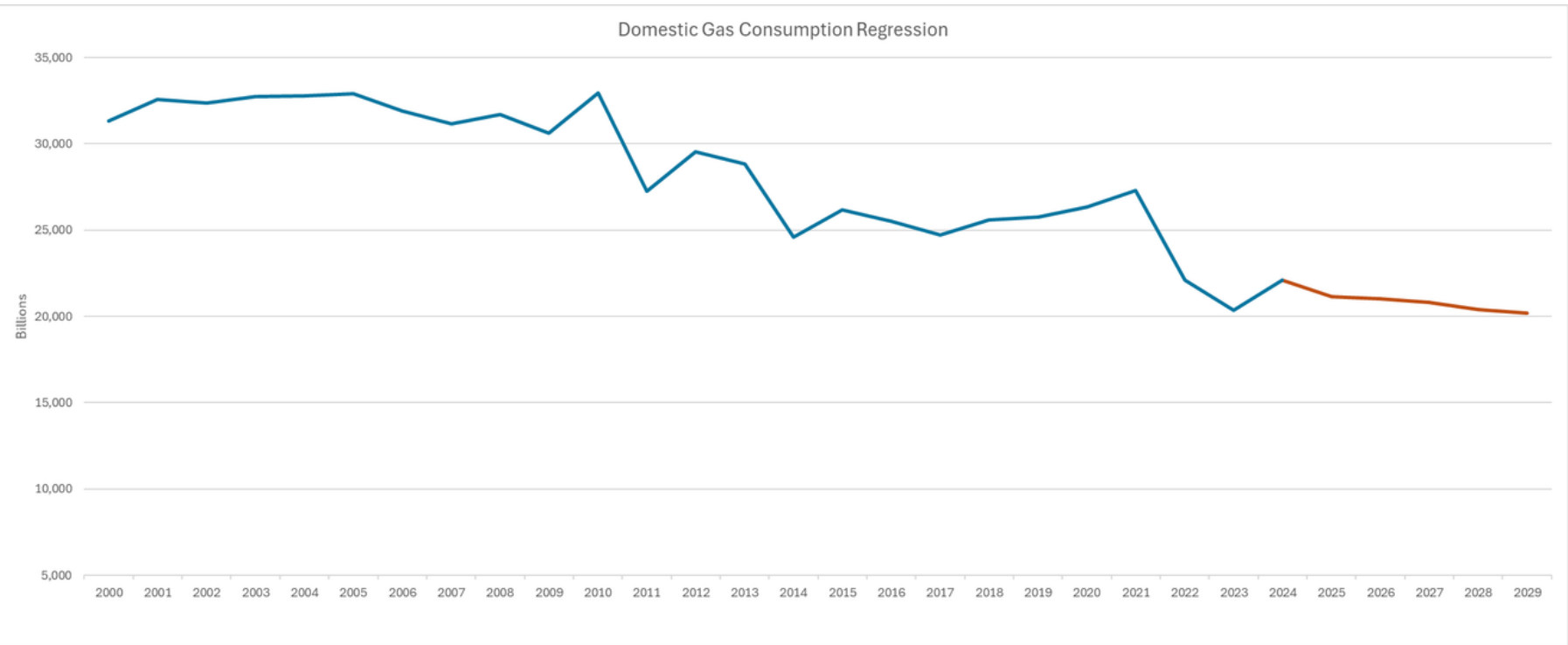
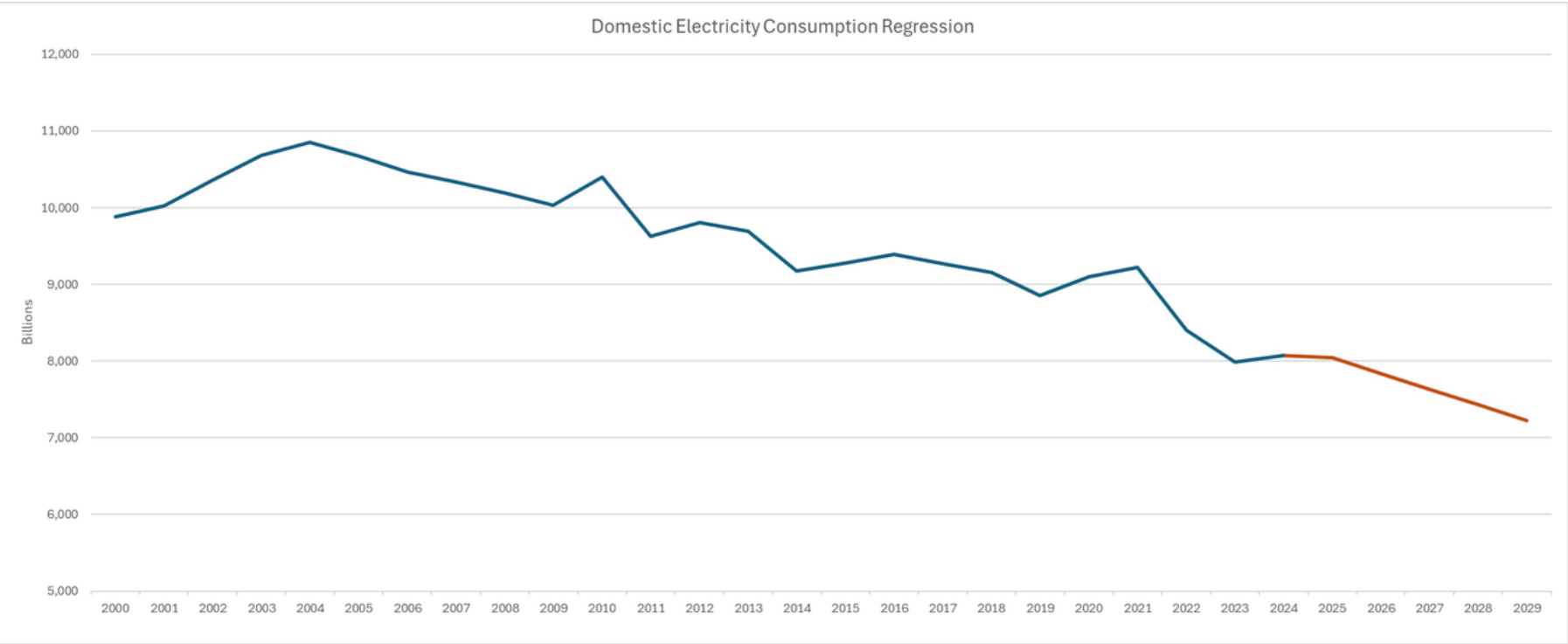
Domestic Electricity Consumption Model

- R-Square
- Variables
- P-values
- Biggest Movers

Domestic Gas Consumption Model

- R-square
- Variables
- P-values
- Biggest Movers

Forecast - Scenario 1

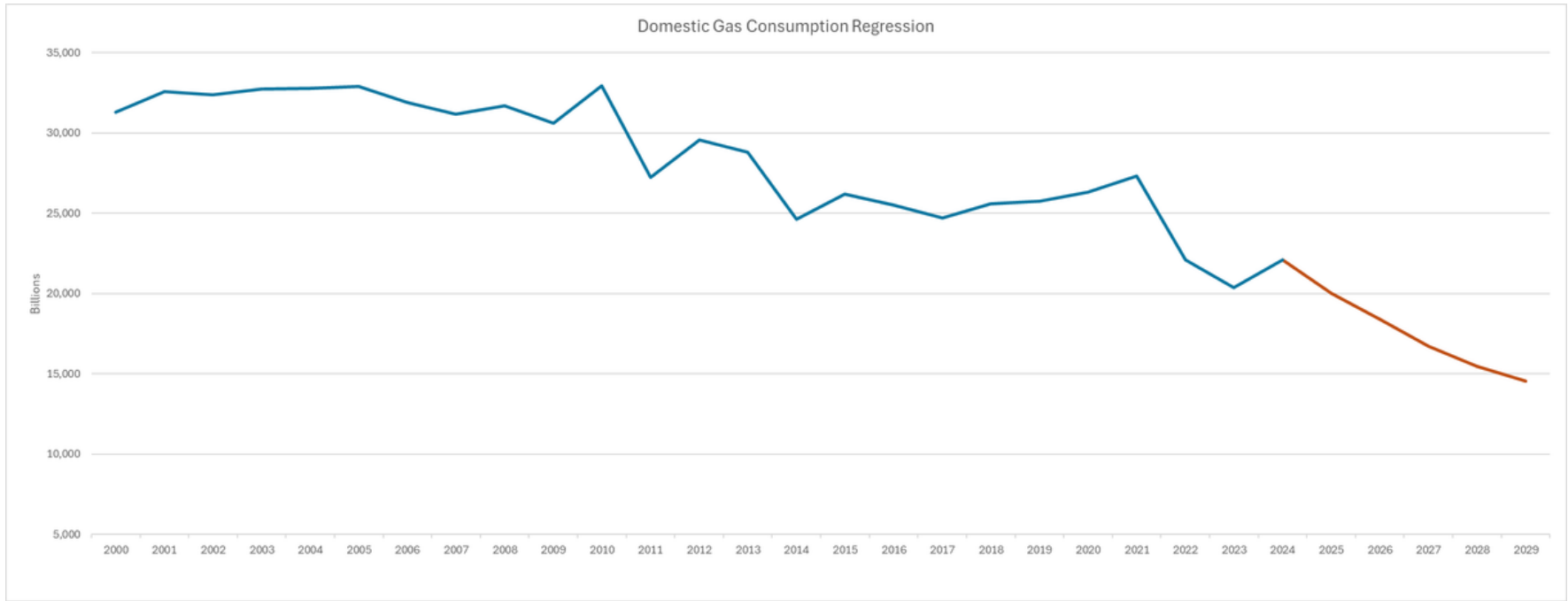
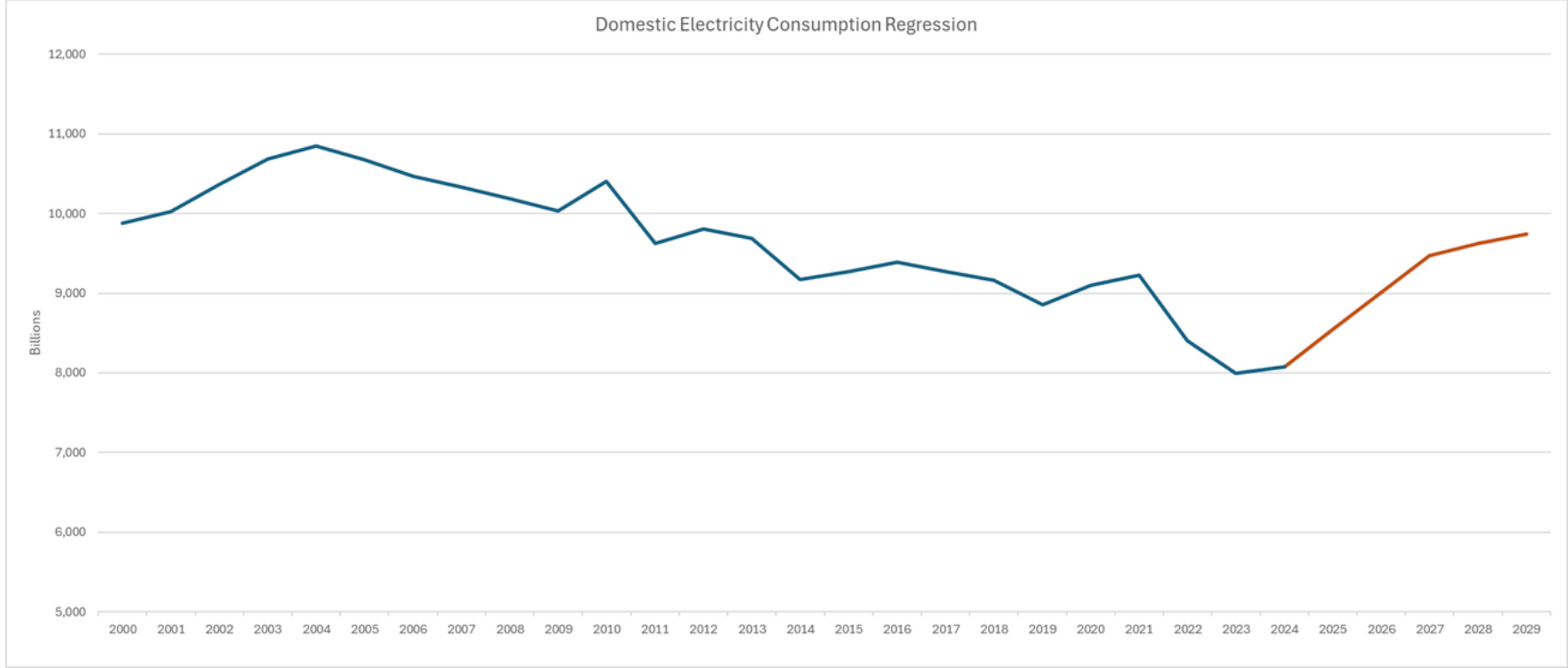


- Electricity prices carry on an upward trend

- Reliance on gas continues



Forecast - Scenario 2



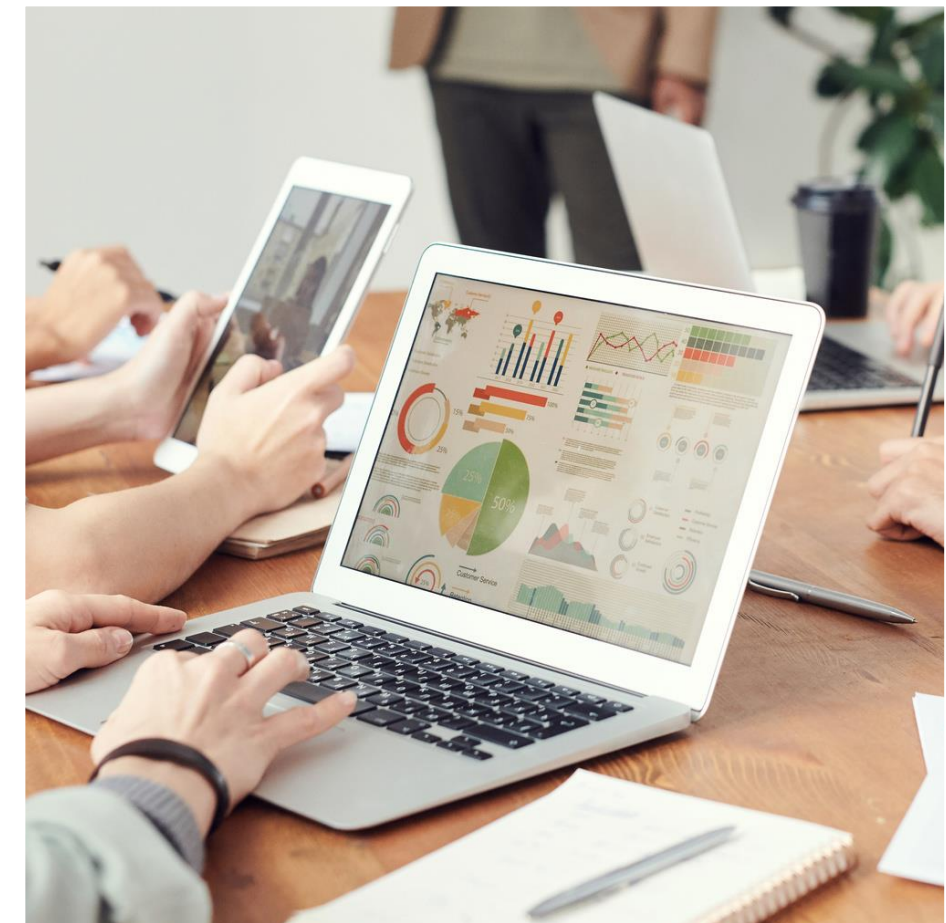
- If electricity prices decrease to pre-covid levels
- This will accelerate our reduction of reliance on gas



Recommendations

– ABOLISH MARGINAL PRICING

Documentation





Thank
You